

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL ENGINEERING

Mini Test August-2024

Course: B.Tech	Semester III	Branch: Electrical Engg.
Subject Name : Electrical Machines-I		Subject Code: EE-212
Max Marks: 10		

Q.No.	Question	Marks
1.	A 3- phase, 4-wire system supplies power load at 400 V and lighting load at 230 V. If the lamps in use require 70, 84 and 33 A in each of the three lines, what should be the current in the neutral wire? If a 3-phase motor is now started, taking 200 A from the line at a power factor of 0.2, what would be the current in each line and the neutral current? Find also the total power supplied to the lamps and the motor.	05
2.	Two wattmeters are used to measure power and power factor of a 3-phase delta connected load having a 25 ohms resistor in series with a 15 mH inductor in each phase. The load is supplied from a 415 V, 50 Hz mains. Determine the readings of the wattmeter.	05